

Problem Statement

Team Gambia — TRI AI Saturdays Cohort 10 — Agricultural Extension RAG: Smart Retrieval for Farmers

The Problem

Retrieval-Augmented Generation (RAG) systems answer questions by first retrieving relevant source documents, then having a language model read them to compose an answer. In practice, the quality of the final answer is capped by the quality of retrieval: if the correct document is never surfaced, no downstream model can produce a correct answer from it, regardless of its size or capability.

This problem is especially high-stakes in agricultural extension services for smallholder farmers in Sub-Saharan Africa. A farmer asking "why are my maize leaves turning yellow?" needs the retrieval system to distinguish between multiple plausible-sounding but distinct causes — nitrogen deficiency, a viral disease, or a fungal infection — each requiring a different, sometimes incompatible, response. Retrieving the wrong factsheet does not just fail to help; it can actively mislead a farmer into an intervention that wastes money or damages the crop.

Why Existing Approaches Fall Short

- **Vocabulary mismatch:** farmers phrase questions colloquially ("matooke wilt") while extension factsheets use formal or scientific terminology ("Banana Bacterial Wilt / *Xanthomonas campestris*"), so keyword-matching retrieval systems frequently fail to connect the two.
- **Hard negatives:** many candidate documents are topically adjacent but factually wrong for a given query (e.g. a potassium-deficiency factsheet retrieved for a nitrogen-deficiency question) — naive lexical or even semantic retrieval can be fooled by shared vocabulary or topic without distinguishing the specific, correct answer.
- **Graded relevance:** a knowledge base typically contains several documents of varying relevance to a single question, so a usable system must rank, not just classify, its candidates.

Our Objective

Build and rigorously evaluate a retrieval engine that ranks a 695-document agricultural extension knowledge base against 200+ real farmer questions, such that the top-5 ranked results maximize nDCG@5 (Normalized Discounted Cumulative Gain), the metric used both for our own local evaluation and for the Kaggle competition this project is built against.

Target Users

- **Smallholder farmers** in Sub-Saharan Africa seeking rapid, accurate answers to crop management questions, often via a mobile-based plant-clinic or chatbot interface.
- **Agricultural extension officers** (e.g. MAAIF "plant doctors" in Uganda) who could use an accurate retrieval layer to speed up their own diagnosis and response process.

- **Downstream RAG/LLM systems** that depend on this retrieval stage to ground their generated answers in accurate source material.

Success Criteria

A retrieval pipeline that measurably and reproducibly outperforms a standard TF-IDF baseline on nDCG@5, validated both on a held-out local evaluation split and on the competition's independent public and private test sets, with every design decision empirically justified against real labeled data rather than assumed from architectural intuition alone.